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CS 170 Midterm 1

Write in the following boxes clearly and then double check.

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Exam Room :
- ● Pimentel 1
- ○ Wheeler 20
- ○ Other (Specify):

Name of student to your left : Aarush

Name of student to your right : N/A

- **The exam will last 110 minutes.**
- The exam has 4 questions, each worth 25 points, plus a challenging bonus question worth only 1 point.
- Answer each question in the box provided. If you need more space, continue on the reverse side of the page and clearly state that you are doing so in the box.
- Good luck!

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1 [25 points] Mining

Deep in a mine, a miner is trying to choose which minerals to carry back out. There are n different minerals, each with some quantity q_i (measured in pounds) available, and some value v_i per pound. The miner wants to bring back minerals with the maximum possible total value, but can only carry 100 pounds back up. In this problem, you will design an efficient algorithm that computes the maximum value of minerals they can carry subject to this constraint.

For each mineral i, the miner may take any fraction of the available q_i pounds, subject to the overall 100 pound maximum. That is, there is no restriction to integer values, and you may analyze running time assuming each real number arithmetic operation takes one timestep.

(a) Give an algorithm to maximize the miner's value, and analyze the runtime.

One algorithm to maximize the miners value is:
Steps: Track current weight w, and make it start at 0. Sort all minerals in descending order, with respect to the minerals value v_i, such that 0 is the highest value mineral and n is the lowest value mineral.
Take the first non-zero mineral i. If $q_i \geq 100 - w$, take $100-w$ lbs of i. Otherwise take q_i lbs of mineral i, add q_i to w, and move to mineral $i+1$.

(b) Prove correctness using an exchange argument.

Assume there exists an optimal solution, s, that is not the greedy solution, g. At the first point where s differs from g, s takes mineral i', and g takes mineral i. $v_i \geq v_{i'}$, since i is the highest value mineral available. Hence the reward generated by s would have been greater by taking i instead of i', so the reward of g is greater than the reward of s, so s is not the optimal solution. Hence, g is the optimal solution by contradiction.

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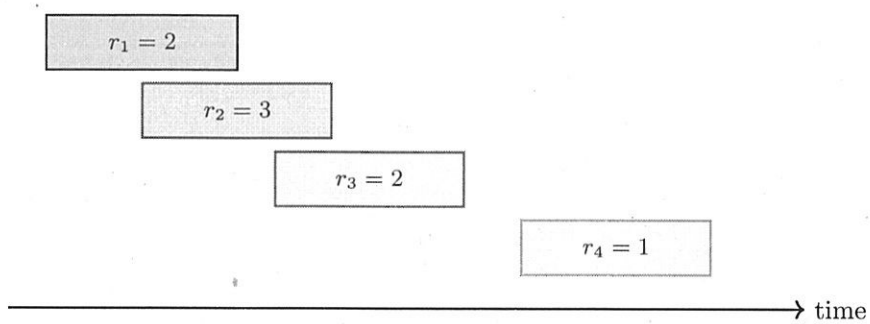
2 [25 points] Scheduling

There are n jobs to be run on a computer. Each job j starts at some time s_j, runs for exactly one hour, and gives a reward payment r_j if it finishes. Two overlapping jobs cannot both be run. The goal of this question is to design an efficient dynamic programming algorithm to find the maximum total reward obtainable from any set of non-overlapping jobs. It is sufficient to describe your **subproblem**, **recurrence**, and **base case**, and **analyze your runtime**.

You may assume the jobs $j = 1, \ldots, n$ are given in increasing order of start time $s_1 < \cdots < s_n$.

For example, for the $n = 4$ jobs drawn below, the maximum reward $r_1 + r_3 + r_4 = 5$ is given by running jobs $1, 3, 4$.



Subproblem:
$R(j)$ is the maximum reward possible by scheduling jobs 1 through j including job j.
$R(j) = \max_{i<j} (R(i) + r_j \text{ if } s_i + 60 < s_j)$

Base Case: $R(0) = 0$, since no jobs generate no reward.
Runtime: Filling entry $R(j)$ requires scanning all $i<j$, so it taken $O(n)$ time to fill 1 entry, and there are n entries to fill, so the algorithm takes $O(n^2)$ time.

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3 [25 points] Shortest Path

You are given a weighted directed graph $G = (V, E)$ in which one edge has weight -1, and all other edges have weight $+1$. You want to find the shortest path from a given source vertex s to a target vertex t.

(a) Give a $O(|V| + |E|)$ time algorithm for this problem, which takes as input the graph $G = (V, E)$, vertices s, t, and the unique edge (u, v) with weight -1. Your algorithm should use breadth-first search.

Conduct breadth first search, tracking the path from s to every visited node, to find a path to t of distance d. Store this path and mark t as unvisited. If (u,v) is one of the edges added on the next step of BFS, and v=t, then the shortest path to t is the shortest path to u found via BFS, and the path from u-v. Otherwise, the shortest path to t is the original path found via normal BFS.

(b) Justify the correctness of your algorithm.

If the path via BFS from s-t has length d, then the true shortest path can have length d or d+1.
If the shortest path has length d it will be found by regular BFS.
If the shortest path has length d+1, then the last connection must be (u,v), since otherwise a shorter length d path exists to t. Hence by checking these two paths, we find the true shortest path.

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4 [25 points] Decimal to Binary

Given the n-digit decimal representation of a positive integer, converting it into binary in the natural way takes $O(n^2)$ steps. In this problem, you will design a more efficient divide-and-conquer algorithm for converting decimal to binary.

Let n be even. Then any n-digit integer can be written as $a \cdot 10^{n/2} + b$ where a and b are both $n/2$-digit integers, e.g. $1342 = 13 \cdot 10^2 + 42$.

(a) Give a divide-and-conquer algorithm to convert an n-digit number to binary.

For an n digit number x, take a to be the first $\frac{n}{2}$ digits and b to be the last $\frac{n}{2}$ digits. x in binary is a in binary $\cdot 10^{\frac{n}{2}}$ in binary + b in binary. This process can then be repeated on a, b, and $10^{\frac{n}{2}}$ until they are single digit numbers. This can be converted to binary in constant time.

Base Case: x is single digit: Convert to binary in constant time using normal method.

Recursive Case: x is not single digit: $x = a \cdot 10^{\frac{n}{2}} + b$ convert a, $10^{\frac{n}{2}}$ and b to binary recursively.

(b) Argue that the running time of your algorithm in part (a) is $O(n^{\log_2(3)} \cdot \log(n))$.

It may be helpful for you to recall that the Karatsuba algorithm for integer multiplication takes $O(n^{\log_2(3)})$ steps to multiply two n-bit integers given in their binary representations.

And, as in Karatsuba, you may assume for simplicity for this question that n is a power of 2.

Note: there is a trick to improving the running time to $O(n^{\log_2(3)})$, so your runtime could be faster than what the question asks for.

Recurrence Relation:

$T(n) = 3T(\frac{n}{2}) + O(n^{\log_2 3})$, since each problem is split into 3 $\frac{n}{2}$ subproblems and multiplying together 2 n bit numbers takes $O(n^{\log_2 3})$ time.

By the master theorem, $a = b^d$, since $3 = 2^{\log_2 3} = 3$, so the runtime is $O(n^d \log n)$ or $O(n^{\log_2 3} \log n)$

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5 [1 points] Bonus Question

Attempt this question only if you have time to spare and enjoy challenging problems. Please note that it is worth only 1 point!

You are given an undirected weighted graph $G = (E, V)$, with non-negative weights $w : E \to Z$. You are also given a disjoint set F of zero-weight "free" edges and an integer k. Give an efficient algorithm to find an MST which uses at most k free edges.

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Assumptions

(0 points) Congratulations! You're done with the exam. Were there any ambiguities in the exam? If so, record the ambiguity and the assumptions that you made in the box below, as well as possible alternate interpretations and their associated answers.

For example, "If question X is asking about A, then my answer is Y. But, if the question is about B, then my answer is Z." If you submit a regrade request regarding any ambiguities on the exam, we will only consider the request if you have correctly noted an ambiguity in the box below.

I assumed s_j was in minutes, so $s_j + 60$ is the completion time of a job.
I assumed BFS was implemented in a way that tracked not only distance, but also the path.
I assumed v_j and

Comments, Concerns, etc.

(0 points) Did you notice anything suspicious during the exam? Please note in the box below. Where were you sitting, and what did you notice?

(0 points) Any other comments, concerns, qualms, queries, or suggestions?

(0 points) 100 penguins versus one CS 170 student. Who would win? Justify your answer.

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